Practical 5

Aim: Implementation of KNN using sklearn

Source-Code:

```
from sklearn.neighbors import KNeighborsClassifier
from sklearn.model_selection import train_test_split
from sklearn.datasets import load_iris
import random

data_iris = load_iris()
label_target = data_iris.target_names
print()

print("Sample Data from Iris Dataset")
print("*"*30)
for i in range(10):
  rn = random.randint(0,120)
  print(data_iris.data[rn],"===>",label_target[data_iris.target[rn]])

X = data_iris.data
y = data_iris.target
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size =
0.3,random_state=1)

print("The Training dataset length: ",len(X_train))
print("The Testing dataset length: ",len(X_test))
try:
  nn = int(input("Enter number of neighbors :"))
  knn = KNeighborsClassifier(nn)
  knn.fit(X_train, y_train)
  # to display the score
  print("The Score is :",knn.score(X_test, y_test))
  # To get test data from the user
  test_data = input("Enter Test Data :").split(",")
  for i in range(len(test_data)):
    test_data[i] = float(test_data[i])
  print()
  v = knn.predict([test_data])
  print("Predicted output is :",label_target[v])
except:
  print("Please supply valid input.....")
```

Output:

```
Sample Data from Iris Dataset
*****************************
[5.  3.4 1.5 0.2] ===> setosa
[7.7 3.8 6.7 2.2] ===> virginica
[5.4 3.  4.5 1.5] ===> versicolor
[7.  3.2 4.7 1.4] ===> versicolor
[7.7 3.8 6.7 2.2] ===> virginica
[6.5 3.  5.8 2.2] ===> virginica
[6.  2.9 4.5 1.5] ===> versicolor
[7.7 3.8 6.7 2.2] ===> virginica
[6.5 2.8 4.6 1.5] ===> versicolor
[5.3 3.7 1.5 0.2] ===> setosa
The Training dataset length:  105
The Testing dataset length:  45
Enter number of neighbors :10
The Score is : 0.9777777777777777
Enter Test Data :2.1,2,3.0,1

Predicted output is : ['setosa']
```